

SAMMYAG SOLANKI

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LinkedIn | GitHub

Education

Vishwakarma Institute of Technology (VIT Pune)

B.Tech. in Electronics & Telecommunication Engineering

2028

CGPA: 8.62/10

Technical Skills

Languages : Python, C, C++, JavaScript, TypeScript, SQL

Machine Learning / Data : XGBoost, Pandas, NumPy, Scikit-learn, Matplotlib

Computer Vision : OpenCV, YOLO

Web / Full Stack : React, Vite, Tailwind CSS, Node.js, Express, Socket.IO, Streamlit

Backend / Cloud / Tools : Firebase Authentication, Firestore, Git, GitHub, VS Code

Projects

IPL Auction Pro

React, TypeScript, Node.js, Socket.IO, Firebase, Gemini API | GitHub

- Developed a real-time multiplayer IPL auction simulator with private rooms, synchronized bidding, live budgets, and squad management.
- Implemented Open, Blind, Draft, and Mega auction formats with automated turn management, player allocation, and state synchronization.
- Integrated Firebase Authentication and Firestore with 127 IPL players, plus Gemini-based scouting and post-auction squad analysis.
- Deployed the full-stack application on Render with a responsive React interface and Node.js/Express backend.

FIFA World Cup 2026 Forecast Model

Python, XGBoost, Streamlit, Monte Carlo Simulation | GitHub

- Built a football forecasting pipeline using 7,326 international matches and 14 engineered predictive features.
- Combined dynamic Elo ratings, recent form, head-to-head statistics, and XGBoost multiclass classification for match-outcome prediction.
- Applied temporal validation, Platt calibration, and historical World Cup backtesting; achieved 60.18% validation accuracy with 0.8760 log loss.
- Developed a Streamlit dashboard and Monte Carlo engine performing 100,000 simulations for tournament-level forecasting.

GeoRush

React, TypeScript, Socket.IO, Express, Google Maps Platform | GitHub

- Developed and deployed a geography-guessing game with single-player, real-time multiplayer, 1v1, Country Streak, Time Attack, and Daily Challenge modes.
- Built a server-authoritative architecture using Node.js, Express, and Socket.IO for rooms, synchronized state, timers, scoring, and round progression.
- Integrated Google Maps JavaScript API and Street View with Haversine-distance scoring, interactive guess maps, and geographic result analysis.
- Implemented XP, levels, achievements, challenge-a-friend links, an Explorer Field Guide, and customization; deployed on Google Cloud Run.

Football Video Analyzer (In Progress)

Python, OpenCV, YOLO, Computer Vision

- Developing a computer-vision pipeline for football match footage using YOLO-based player and ball detection.
- Implementing multi-object tracking and persistent player identities across video frames for movement and match-analysis features.
- Building an annotated-video processing workflow as the foundation for player-level sports analytics.

Publication

Dual-Axis Sun Tracking Solar Panel

IEEE Xplore

IEEE-published research work on a dual-axis solar tracking system.

RELEVANT COURSEWORK

Data Structures & Algorithms | Object-Oriented Programming | Operating Systems | Design & Analysis of Algorithms | Computer Networks | Machine Learning & Computer Vision | Embedded System Design